CHEMICAL THERMODYNAMICS: MASTER TEXTBOOK

1. Fundamental Principles

System and Surroundings

Thermodynamics studies energy changes in chemical processes. A **System** is the part of the universe under study, while the **Surroundings** is everything else.

2. Laws of Thermodynamics

First Law (Conservation of Energy)

Energy cannot be created or destroyed, only transformed.

$$\Delta U = q + w$$

Second Law (Entropy)

For any spontaneous process, the total entropy of the universe always increases ($\Delta S_{univ} > 0$).

3. Enthalpy and Gibbs Free Energy

Predicting Spontaneity

Enthalpy (H): Total heat content of a system.

Gibbs Free Energy (G): The criteria for spontaneity.

$$\Delta G = \Delta H - T\Delta S$$

If $\Delta G < 0$, the reaction is spontaneous. If $\Delta G > 0$, it is non-spontaneous. If $\Delta G = 0$, the system is at equilibrium.

4. Thermodynamic Processes

- **Isothermal:** Temperature remains constant ($\Delta T=0$).
- **Adiabatic:** No heat exchange ($q=0$).
- **Isobaric:** Pressure remains constant ($\Delta P=0$).
- **Isochoric:** Volume remains constant ($\Delta V=0$).